

Laporan Jobsheet 1: Setup Project Matkul Pemograman Berbasis Network

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D. Langkah Kerja Praktikum

Langkah 1 – Pengecekan Lingkungan

```
PS C:\Users\ahmad> node -v
v24.13.1
PS C:\Users\ahmad> npm -v
11.8.0
PS C:\Users\ahmad> git -v
git version 2.47.0.windows.2
PS C:\Users\ahmad> |
```

Langkah 2 – Membuat Project Next.js

```
PS D:\pemograman_framework\Code\praktikum> npx create-next-app@13.4.7
Need to install the following packages:
create-next-app@13.4.7
Ok to proceed? (y) y
/ What is your project named? ... my-app
/ Would you like to use TypeScript with this project? ... No / Yes
/ Would you like to use ESLint with this project? ... Yes
/ Would you like to use Tailwind CSS with this project? ... No / Yes
/ Would you like to use 'src/' directory with this project? ... No / Yes
/ Use App Router (recommended)? ... No / Yes
/ Would you like to customize the default import alias? ... No / Yes
Creating a new Next.js app in D:\pemograman_framework\Code\praktikum\my-app.

Using npm.

Initializing project with template: default

Installing dependencies:
- react
- react-dom
- next
- eslint
- eslint-config-next

added 335 packages, and audited 336 packages in 2m
138 packages are looking for funding
  run `npm fund` for details
16 vulnerabilities (3 moderate, 13 high)
To address all issues, run:
  npm audit fix

Run `npm audit` for details.
Success! Created my-app at D:\pemograman_framework\Code\praktikum\my-app
A new version of 'create-next-app' is available!
You can update by running: npm i -g create-next-app
PS D:\pemograman_framework\Code\praktikum> |
```

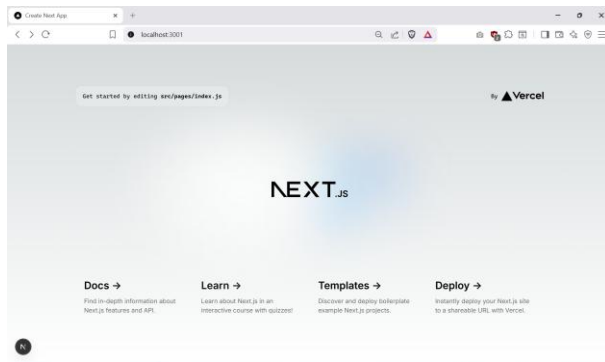
Langkah 3 – Menjalankan Server Development

```
PS D:\pemograman_framework\Code\praktikum\my-app> npm run dev

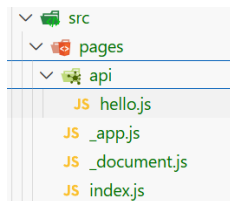
> my-app@0.1.0 dev
> next dev

✓ Console Ninja extension is connected to Next.js, see https://tinyurl.com/2vt8jxzw
⚠ Port 3000 is in use by process 11256, using available port 3001 instead.
▲ Next.js 16.1.6 (TurboPack)
- Local:      http://localhost:3001
- Network:    http://172.16.0.2:3001

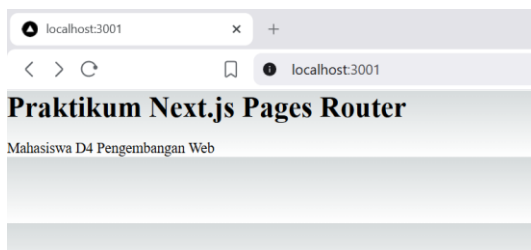
✓ Starting...
✓ Ready in 552ms
GET / 200 in 1131ms (compile: 1044ms, render: 88ms)
```



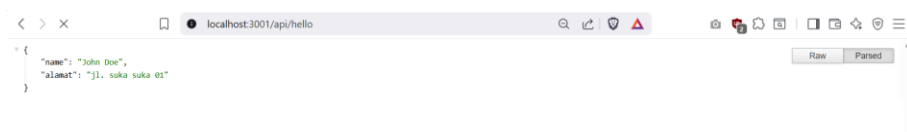
Langkah 4 – Mengenal Struktur Folder



Langkah 5 – Modifikasi Halaman Utama



Langkah 6 – Modifikasi API



Langkah 7 – Modifikasi Background

Praktikum Next.js Pages Router

Mahasiswa D4 Pengembangan Web

Langkah 8 – Setup ext pada VSCode (opsional)

E. Tugas Praktikum

Tugas 1 (Wajib)

- Buat halaman baru about.js di folder pages.
- Tampilkan:
 - o Nama Mahasiswa
 - o NIM
 - o Program Studi

Jawaban:

```
abouttsx U X
Code > praktikum > my-app > src > pages > abouttsx > Home
6  const inter = Inter({ subsets: ['latin'] })
7
Windsurf: Refactor | Explain | Generate JSDoc | X
8  export default function Home() {
9    return (
10     <div>
11       <h1>About Page</h1> <br />
12       <p>Ini adalah halaman About</p>
13       <p>Nama: Ahmad Dzul Fadhli Hannan</p>
14       <p>NIM: 2341720106</p>
15       <p>Program Studi: D4 Teknik Informatika</p>
16     </div>
17   )
18 }
19
```

localhost:3001/about × +

< > ↻ 📖 ⓘ localhost:3001/about

About Page

Ini adalah halaman About

Nama: Ahmad Dzul Fadhli Hannan

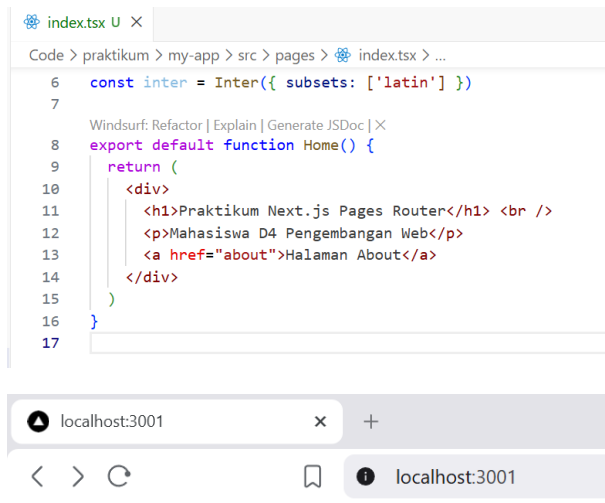
NIM: 2341720106

Program Studi: D4 Teknik Informatika

Tugas 2 (Pengayaan)

- Tambahkan minimal 1 link navigasi dari halaman utama ke halaman about.

Jawaban:



The image shows a code editor window with a file named `index.tsx`. The code defines an `Inter` component and a `Home` function. The `Home` function returns a JSX element containing an `h1`, a `p`, and an `a` tag with a link to `about`. Below the code editor, a web browser window is shown with the address `localhost:3001`. The browser displays the rendered output of the `Home` function, which includes the text "Praktikum Next.js Pages Router", "Mahasiswa D4 Pengembangan Web", and a link "Halaman About".

```
6  const inter = Inter({ subsets: ['latin'] })
7
8  export default function Home() {
9    return (
10     <div>
11       <h1>Praktikum Next.js Pages Router</h1> <br />
12       <p>Mahasiswa D4 Pengembangan Web</p>
13       <a href="about">Halaman About</a>
14     </div>
15   )
16 }
17
```

Praktikum Next.js Pages Router

Mahasiswa D4 Pengembangan Web

[Halaman About](#)

F. Pertanyaan Refleksi

1. Mengapa Pages Router disebut sebagai routing berbasis file?

Karena navigasi halaman ditentukan secara otomatis oleh struktur folder dan nama file di dalam direktori pages tanpa perlu konfigurasi manual.

2. Apa perbedaan Next.js dengan React standar (CRA)?

Next.js mendukung *Server-Side Rendering* (SSR) dan memiliki fitur bawaan seperti routing, sedangkan React standar (CRA) menggunakan *Client-Side Rendering* (CSR) dan memerlukan library tambahan untuk navigasi.

3. Apa fungsi perintah `npm run dev`?

Untuk menjalankan server pengembangan lokal yang memungkinkan untuk melihat perubahan kode secara langsung

4. Apa perbedaan `npm run dev` dan `run build` ?

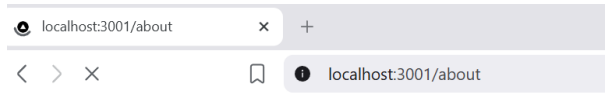
npm run dev digunakan untuk proses pengodean / pengembangan, npm run build digunakan untuk mengoptimasi dan mengonversi aplikasi menjadi file siap produksi / di deploy.

Laporan Jobsheet 2: PageDanLayout Matkul Pemograman Berbasis Network

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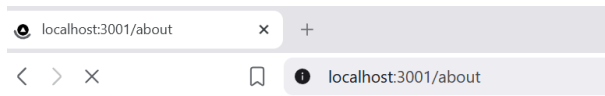
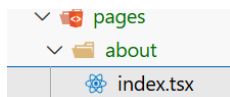
D. Langkah Praktikum

1. Routing Dasar (Static Routing)



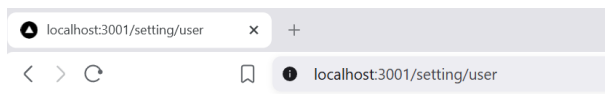
Ini Adalah Halaman About

2. Routing Menggunakan Folder

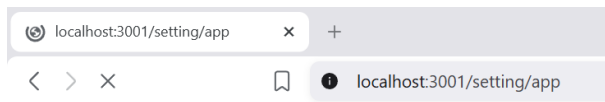


Ini Adalah Halaman About

3. Nested Routing

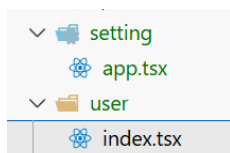


User Setting Page

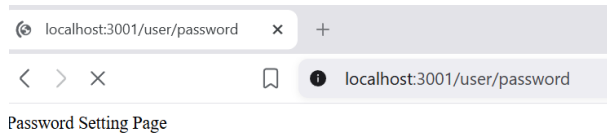


App Setting Page

Modifikasi

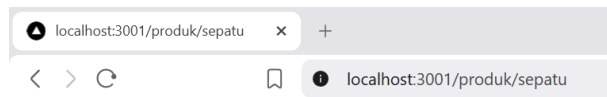


Nested Lebih Dalam



4. Dynamic Routing

Console Log

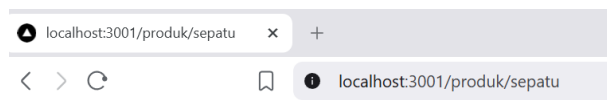


Halaman Produk

Produk:

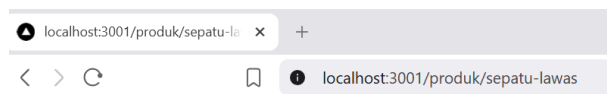


Statis Router



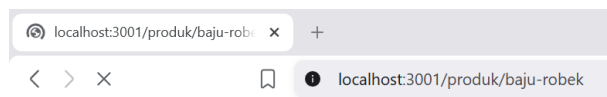
Halaman Produk

Produk: sepatu



Halaman Produk

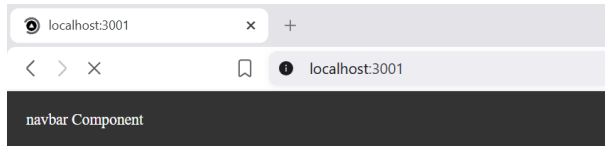
Produk: sepatu-lawas



Halaman Produk

Produk: baju-robek

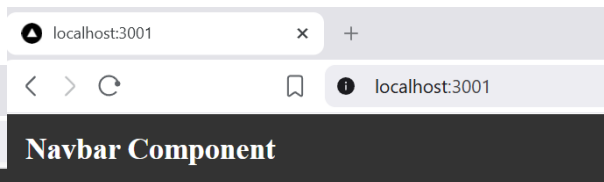
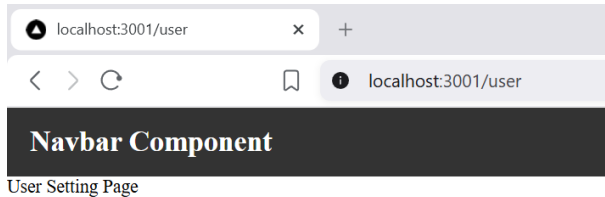
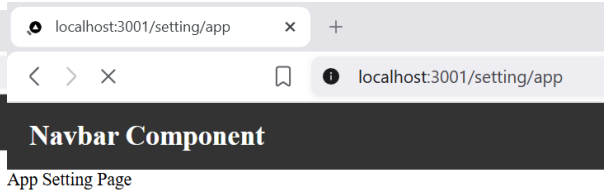
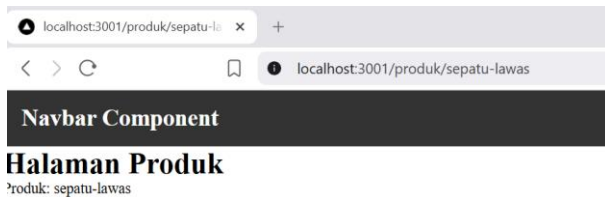
5. Membuat Komponen Navbar



Praktikum Next.js Pages Router

Mahasiswa D4 Pengembangan Web
Halaman About

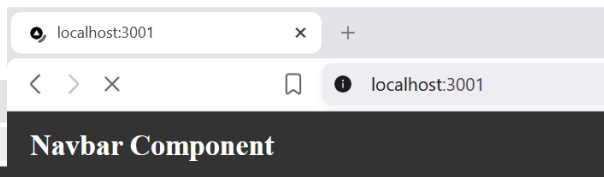
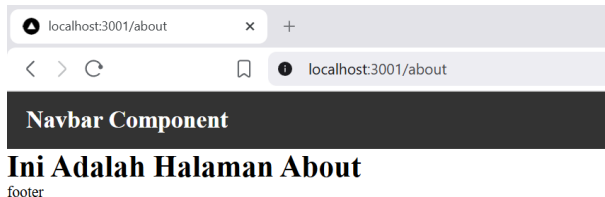
Terapkan ke semua page



Praktikum Next.js Pages Router

Mahasiswa D4 Pengembangan Web
Halaman About

6. Membuat Layout Global (App Shell)



Praktikum Next.js Pages Router

Mahasiswa D4 Pengembangan Web
Halaman About
footer

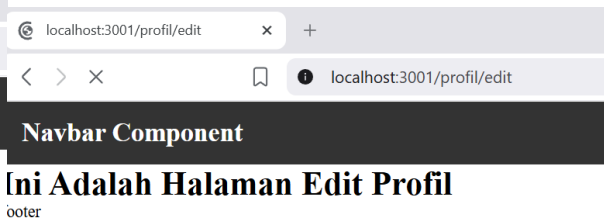
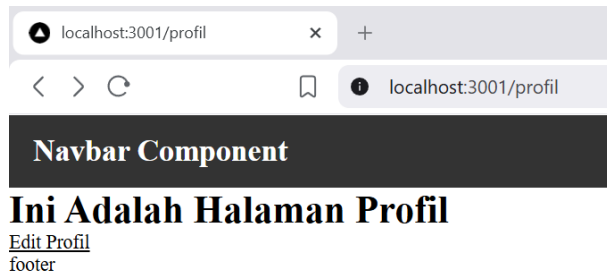
E. Tugas Praktikum

Tugas 1 – Routing

1. Buat halaman:

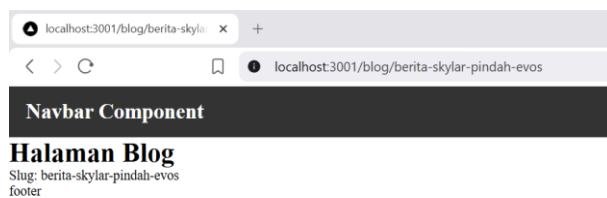
- o /profile
- o /profile/edit

2. Pastikan routing berjalan tanpa error



Tugas 2 – Dynamic Routing

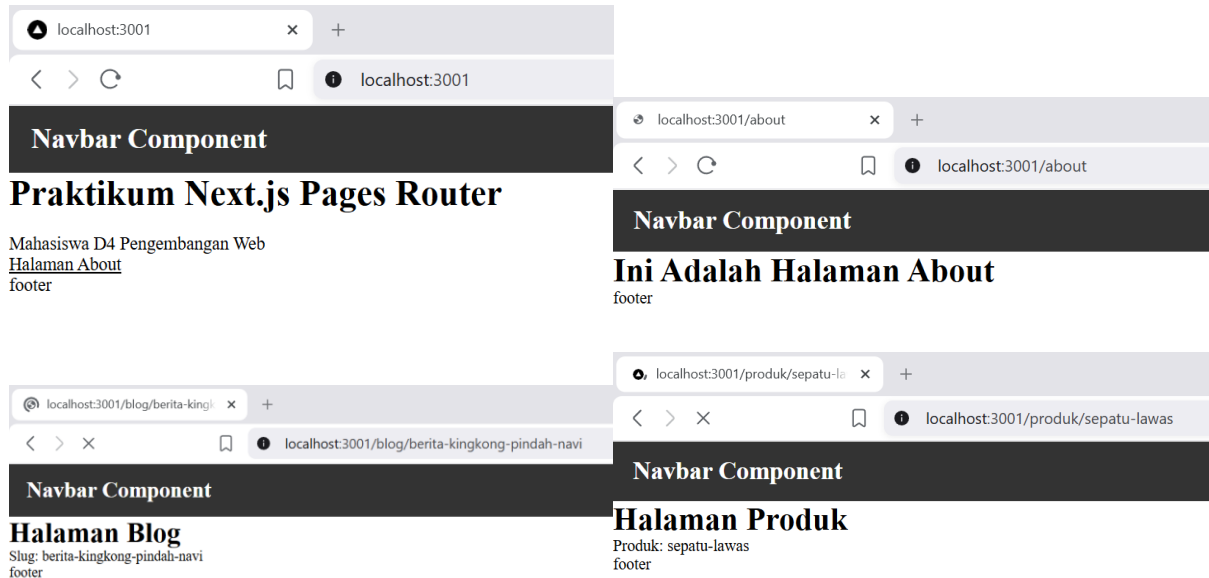
1. Buat routing:
2. `/blog/[slug]`
3. Tampilkan nilai slug di halaman



Tugas 3 – Layout

1. Tambahkan Footer pada AppShell
2. Footer tampil di semua halaman

```
Code > praktikum > my-app > src > components > layouts > Appshell > index.tsx > AppShell
1  import Navbar from "../navbar";
2
3  Windsurf: Refactor | Explain
4  type AppShellProps = {
5    children: React.ReactNode;
6  }
7
8  Windsurf: Refactor | Explain | Generate JSDoc [X]
9  const AppShell = (props: AppShellProps) => {
10    const {children} = props;
11    return (
12      <main>
13        <Navbar />
14        {children}
15        <div>
16          footer
17        </div>
18      </main>
19    );
20  };
21
22 export default AppShell;
```



F. Pertanyaan Refleksi

1. Apa perbedaan routing berbasis file dan routing manual?

Routing berbasis file secara otomatis membuat struktur file di direktori tertentu menjadi URL, sedangkan routing manual harus mendefinisikan setiap jalur dan komponennya secara eksplisit (biasanya ada di file route) di dalam kode program.

2. Mengapa dynamic routing penting dalam aplikasi web?

Karena memungkinkan satu template halaman menangani berbagai data unik berdasarkan parameter URL, seperti ID produk atau nama pengguna, tanpa perlu membuat file terpisah untuk setiap konten.

3. Apa keuntungan menggunakan layout global dibanding memanggil komponen satu per satu?

Karena dapat membungkus seluruh halaman aplikasi secara konsisten (seperti navbar dan footer) tanpa harus memanggil dan menulis ulang kode komponen tersebut di setiap file halaman secara manual.